KON 313
Feedback Control Systems

Stability

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Contents

Introduction

- **Stability** is the **most important** system specification. If a system is unstable, transient and steady-state response characteristics have no meanings.

- An **unstable** system **cannot** be designed for a specific transient response or steady-state error requirement.

- There are many definitions for stability, depending upon the kind of system or the point of view. Here, definitions are limited to linear, time-invariant (**LTI**) systems.

Definition: Using Natural Response Concept

We can **control** the output of a system if the steady-state response consists of **only the forced response**. But, the total response of a system is the sum of the forced and natural responses

$$c(t) = c_{\text{forced}}(t) + c_{\text{natural}}(t)$$

- **Stability.** An LTI system is **stable** if the natural response approaches zero as time approaches infinity.
- **Instability.** An LTI system is **unstable** if the natural response grows without bound as time approaches infinity.
- **Marginal Stability.** An LTI system is **marginally stable** if the natural response neither decays nor grows but remains constant or oscillates as time approaches infinity.

Thus, the definition of stability implies that **only the forced response remains** as the natural response approaches zero.

Definition: Using Natural Response Concept

Disadvantage:

- Looking at the total response, it may be difficult to **separate** the natural response from the forced response.
  - However, we realize that if the **input** is **bounded** and the total response is not approaching infinity as time approaches infinity, then the natural response is obviously not approaching infinity.
  - If the **input** is **unbounded**, we see an unbounded total response, and we **cannot** arrive at any conclusion about the stability of the system; we cannot tell whether the total response is unbounded because the forced response is unbounded or because the natural response is unbounded.

Definition: Regarding the Total Response (BIBO)

This alternate definition is:

- **Stability.** A system is stable if every bounded input yields a bounded output. We call this statement the **bounded-input, bounded-output (BIBO)** definition of stability.
- **Instability.** A system is **unstable** if any **bounded** input yields an **unbounded** output.
- **Marginal Stability.** Using total response, **marginal stability** means that the system is stable for some bounded inputs and unstable for others. Thus, marginally stable systems by the natural response definitions are included as unstable systems under the BIBO definitions.

From the perspective of the time response plot of a physical system, instability is displayed by transients that grow without bound and, consequently, **a total response that does not approach a steady-state value**.

Stability based on the Poles of Closed Loop System

Recall:

- Poles in the **left half-plane** (LHP) yield either pure exponential decay or damped sinusoidal natural responses. These natural responses **decay to zero** as time approaches infinity.
- Poles in the **right half-plane** (RHP) yield either pure exponentially increasing or exponentially increasing sinusoidal natural responses. These natural responses approach **infinity** as time approaches infinity.
- Also, poles of **multiplicity greater than 1 on the imaginary axis** lead to the sum of responses of the form $At^n \cos(\omega t + \phi)$, where $n = 1, 2, \ldots$ where the amplitude approaches **infinity** as time approaches infinity.
- A system that has **imaginary axis poles of multiplicity 1** yields pure sinusoidal **oscillations** as a natural response.

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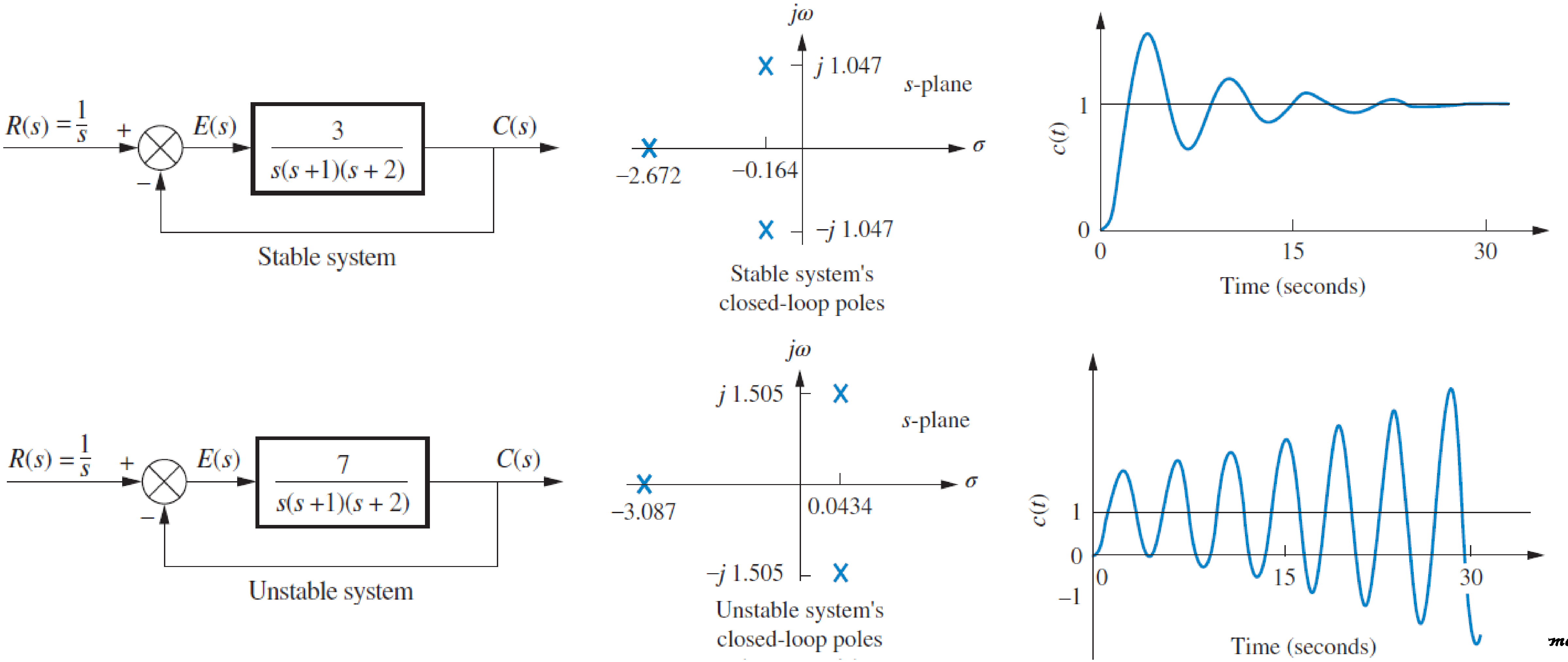
Stability based on the Poles of Closed Loop System

Stability can be determined based on the location of the poles of the closed loop system with respect to the imaginary axis:

- **Stable** systems have closed-loop transfer functions with poles **only in the left half-plane** (poles with a negative real part).
- **Unstable** systems have closed-loop transfer functions with **at least one pole in the right half-plane** (poles with a positive real part) **and/or poles of multiplicity greater than 1 on the imaginary axis**.
- **Marginally stable** systems have closed-loop transfer functions with **only imaginary axis poles of multiplicity 1 and poles in the left half-plane**.

Stability based on the Poles of Closed Loop System

It is not always simple to determine if a closed loop system is stable. Although we know the poles of the forward transfer function, **we do not know the location of the poles of the closed-loop system** without solving for the roots.



Routh-Hurwitz Criterion

- This is a method that yields stability information without having to solve for the closed-loop system poles.
- Using this method, we can tell how many closed-loop system poles are in the left half-plane, in the right halfplane, and on the $j\omega$-axis.
- Notice that we say how many, not where.
- We can find the number of poles in each section of the s-plane, but we cannot find their coordinates.

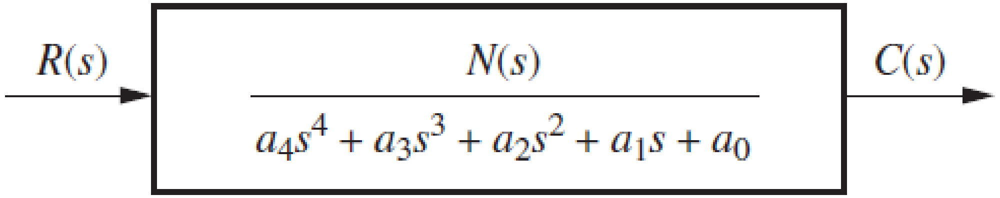
The method requires two steps:

1) Generate a data table called a Routh table
2) Interpret the Routh table to tell how many closed loop system poles are in the left half-plane, the right half-plane, and on the $j\omega$-axis.

Routh-Hurwitz Criterion: Generating Basic Routh Table

Since we are interested in the system poles, we focus our attention on the denominator.

1. Begin by labeling the rows with powers of s from the highest power of the denominator of the closed loop transfer fn. to s^0.
2. Start with the coefficient of the highest power of s in the denominator and list, horizontally in the first row, every other coefficient.
3. In the second row, list horizontally, starting with the next highest power of s, every coefficient that was skipped in the first row.



s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2			
s^1			
s^0			

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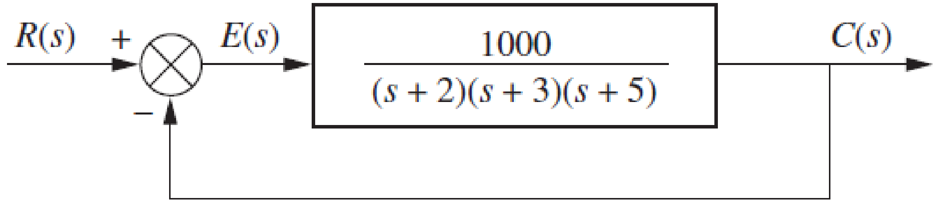
Routh-Hurwitz Criterion: Generating Basic Routh Table

4. The remaining entries are filled in as follows.
   - Each entry is a negative determinant of entries in the previous two rows divided by the entry in the first column directly above the calculated row.
   - The left-hand column of the determinant is always the first column of the previous two rows, and the right-hand column is the elements of the column above and to the right.
   - The table is complete when all the rows are completed down to s^0.

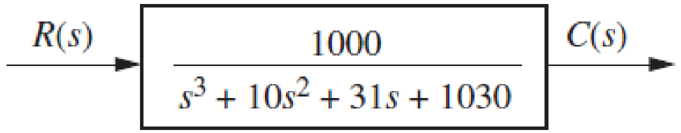
s^4	a_4	a_2	a_0
s^3	a_3	a_1	0
s^2	$\dfrac{-\begin{vmatrix} a_4 & a_2 \\ a_3 & a_1 \end{vmatrix}}{a_3} = b_1$	$\dfrac{-\begin{vmatrix} a_4 & a_0 \\ a_3 & 0 \end{vmatrix}}{a_3} = b_2$	$\dfrac{-\begin{vmatrix} a_4 & 0 \\ a_3 & 0 \end{vmatrix}}{a_3} = 0$
s^1	$\dfrac{-\begin{vmatrix} a_3 & a_1 \\ b_1 & b_2 \end{vmatrix}}{b_1} = c_1$	$\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$	$\dfrac{-\begin{vmatrix} a_3 & 0 \\ b_1 & 0 \end{vmatrix}}{b_1} = 0$
s^0	$\dfrac{-\begin{vmatrix} b_1 & b_2 \\ c_1 & 0 \end{vmatrix}}{c_1} = d_1$	$\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$	$\dfrac{-\begin{vmatrix} b_1 & 0 \\ c_1 & 0 \end{vmatrix}}{c_1} = 0$

Routh-Hurwitz Criterion: Generating Basic Routh Table

Ex. Make the Routh table for the system given below.



Solution. The first step is to find the equivalent closed-loop system because we want to test the denominator of this function, not the given forward transfer function, for pole location.



Routh-Hurwitz criterion will be applied to this denominator.

s^3	1	31	0
s^2	$\cancel{10}\quad 1$	$\cancel{1030}\quad 103$	0
s^1	$\dfrac{-\begin{vmatrix}1 & 31\\ 1 & 103\end{vmatrix}}{1} = -72$	$\dfrac{-\begin{vmatrix}1 & 0\\ 0 & 0\end{vmatrix}}{1} = 0$	$\dfrac{-\begin{vmatrix}1 & 0\\ 1 & 0\end{vmatrix}}{1} = 0$
s^0	$\dfrac{-\begin{vmatrix}1 & 103\\ -72 & 0\end{vmatrix}}{-72} = 103$	$\dfrac{-\begin{vmatrix}1 & 0\\ -72 & 0\end{vmatrix}}{-72} = 0$	$\dfrac{-\begin{vmatrix}1 & 0\\ -72 & 0\end{vmatrix}}{-72} = 0$

Note: For convenience, any row of the Routh table can be multiplied by a positive constant without changing the values of the rows below.

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Routh-Hurwitz Criterion: Interpreting Basic Routh Table

The basic Routh table applies to systems with poles in the left and right half-planes. Systems with imaginary poles and the kind of Routh table that results will be discussed in next slides.

The Routh-Hurwitz criterion declares that the

- number of roots of the polynomial that are in the right half-plane is equal to the number of sign changes in the first column.
- If the closed-loop transfer function has all poles in the left half of the s-plane, the system is stable.
- Thus, a system is stable if there are no sign changes in the first column of the Routh table.

Routh-Hurwitz Table: Special Cases

Two special cases can occur:

1. The Routh table sometimes will have a zero only in the first column of a row.
2. The Routh table sometimes will have an entire row that consists of zeros.

Routh-Hurwitz Table: Special Cases

Special Case-1: Zero Only in the First Column

If the first element of a row is zero, division by zero would be required to form the next row.

1. To avoid this phenomenon, an epsilon, ε, is assigned to replace the zero in the first column.

2. The value ε is then allowed to approach zero from either the positive or the negative side, after which the signs of the entries in the first column can be determined.

Routh-Hurwitz Table: Special Cases

Ex. Determine the stability of the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3}$$

s^5	1	3	5
s^4	2	6	3
s^3	$\cancel{0}$ ϵ	$\frac{7}{2}$	0
s^2	$\frac{6\epsilon - 7}{\epsilon}$	3	0
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	0	0
s^0	3	0	0

Label	**First column**	$\epsilon = +$	$\epsilon = -$
s^5	1	+	+
s^4	2	+	+
s^3	$\cancel{0}$ ϵ	+	−
s^2	$\frac{6\epsilon - 7}{\epsilon}$	−	+
s^1	$\frac{42\epsilon - 49 - 6\epsilon^2}{12\epsilon - 14}$	+	+
s^0	3	+	+

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Routh-Hurwitz Table: Special Cases

Special Case-2: Entire Row is Zero

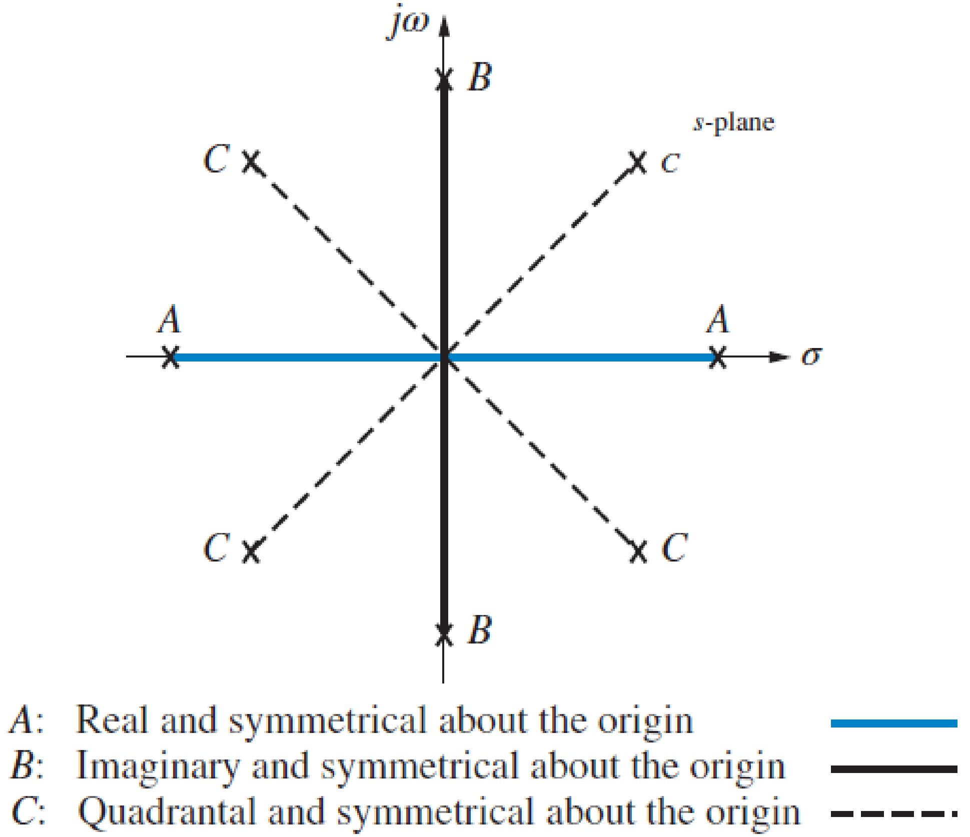
Sometimes while making a Routh table, we find that an entire row consists of zeros. An entire row of zeros will appear in the Routh table when a purely even (i.e. it has only even powers of s) or purely odd polynomial is a factor of the original polynomial.

Even polynomials only have roots that are symmetrical about the origin. This symmetry can occur under three conditions of root position:

(1) The roots are symmetrical and real,

(2) the roots are symmetrical and imaginary, or

(3) the roots are quadrantal.



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Routh-Hurwitz Table: Special Cases

Special Case-2: Entire Row is Zero

Observations:

- Since $j\omega$ roots are symmetric about the origin, If we do not have a row of zeros, we **cannot** possibly have $j\omega$ roots.
- Another characteristic of the Routh table for this case is that **the row previous to the row of zeros** contains the even polynomial that is a factor of the original polynomial.
- Finally, everything from the row containing the even polynomial down to the end of the Routh table is a test of only the even polynomial.

Routh-Hurwitz Table: Special Cases

Special Case-2: Entire Row is Zero

Ex.

Determine the number of right-half-plane poles in the closed-loop transfer function

$$T(s) = \frac{10}{s^5 + 7s^4 + 6s^3 + 42s^2 + 8s + 56}$$

s^5			1			6			8
s^4		$\cancel{7}$	1		$\cancel{42}$	6		$\cancel{56}$	8
s^3	$\cancel{0}$	$\cancel{4}$	1	$\cancel{0}$	$\cancel{12}$	3	$\cancel{0}$	$\cancel{0}$	0
s^2			3			8			0
s^1			$\frac{1}{3}$			0			0
s^0			8			0			0

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Routh-Hurwitz Table: Special Cases

Ex.

PROBLEM: For the transfer function

$$T(s) = \frac{20}{s^8 + s^7 + 12s^6 + 22s^5 + 39s^4 + 59s^3 + 48s^2 + 38s + 20} \tag{6.11}$$

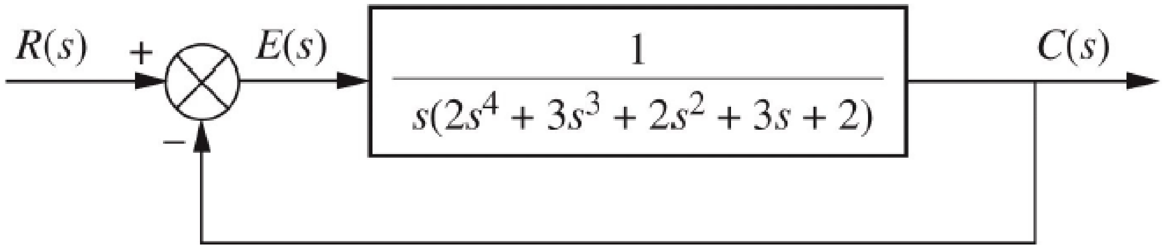
tell how many poles are in the right half-plane, in the left half-plane, and on the $j\omega$-axis.

TABLE 6.8 Routh table for Example 6.5

s^8	1	12	39	48	20
s^7	1	22	59	38	0
s^6	$-\cancel{10}\ -1$	$-\cancel{20}\ -2$	$\cancel{10}\ \ 1$	$\cancel{20}\ \ 2$	0
s^5	$\cancel{20}\ \ 1$	$\cancel{60}\ \ 3$	$\cancel{40}\ \ 2$	0	0
s^4	1	3	2	0	0
s^3	$\cancel{0}\ \ \cancel{4}\ \ 2$	$\cancel{0}\ \ \cancel{6}\ \ 3$	$\cancel{0}\ \ \cancel{0}\ \ 0$	0	0
s^2	$\cancel{\frac{3}{2}}\ \ 3$	$\cancel{2}\ \ 4$	0	0	0
s^1	$\frac{1}{3}$	0	0	0	0
s^0	4	0	0	0	0

Routh-Hurwitz Table: Special Cases

Ex Find the number of poles in the left half-plane, the right half-plane, and on the imaginary-axis for the system of



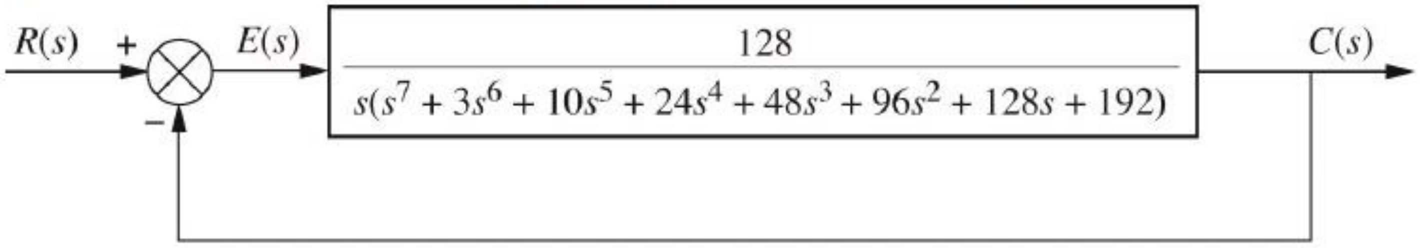
Solution: The closed-loop transfer function is

$$T(s) = \frac{1}{2s^5+3s^4+2s^3+3s^2+2s+1}$$

s^5	2	2	2
s^4	3	3	1
s^3	$\cancel{0}\quad \epsilon$	$\frac{4}{3}$	
s^2	$\frac{3\epsilon-4}{\epsilon}$	1	
s^1	$\frac{12\epsilon-16-3\epsilon^2}{9\epsilon-12}$		
s^0	1		

Routh-Hurwitz Table: Special Cases

Ex. Find the number of poles in the left half-plane, the right half-plane, and on the imaginary-axis for the system



Solution: The closed-loop transfer function for the system is

$$T(s) = \frac{128}{s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128}$$

s^8	1	10	48	128	128
s^7	$\cancel{3}\ \ 1$	$\cancel{24}\ \ 8$	$\cancel{96}\ \ 32$	$\cancel{192}\ \ 64$	
s^6	$\cancel{2}\ \ 1$	$\cancel{16}\ \ 8$	$\cancel{64}\ \ 32$	$\cancel{128}\ \ 64$	
s^5	$\cancel{0}\ \ \cancel{6}\ \ 3$	$\cancel{0}\ \ \cancel{32}\ \ 16$	$\cancel{0}\ \ \cancel{64}\ \ 32$	$\cancel{0}\ \ \cancel{0}\ \ 0$	
s^4	$\cancel{\frac{8}{3}}\ \ 1$	$\cancel{\frac{64}{3}}\ \ 8$	$\cancel{64}\ \ 24$		
s^3	$\cancel{-8}\ -1$	$\cancel{-40}\ -5$			
s^2	$\cancel{3}\ \ 1$	$\cancel{24}\ \ 8$			
s^1	3				
s^0	8				

Routh-Hurwitz Table: Special Cases

Ex.

Find the range of gain, K, for the system given that will cause the system to be stable, unstable, and marginally stable. Assume $K > 0$.

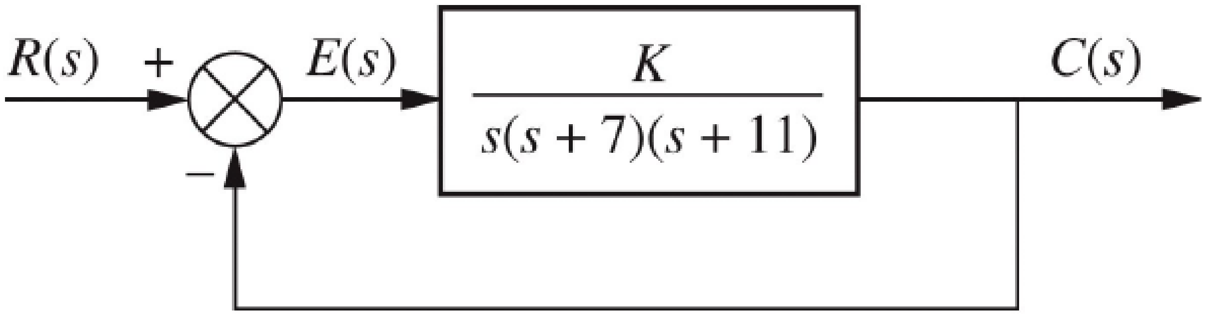


Figure 6.10

$$T(s) = \frac{K}{s^3 + 18s^2 + 77s + K}$$

Solution.

s^3	1	77
s^2	18	K
s^1	$\frac{1386 - K}{18}$	
s^0	K	

All elements in the first column are always positive except the s^1 row. This entry can be positive, zero, or negative, depending upon the value of K.

Routh-Hurwitz Table: Special Cases

Solution Cont.

- If $K < 1386$, all terms in the first column will be positive, and since there are no sign changes, the system will have three poles in the LHP and be stable.
- If $K > 1386$, the s^1 term in the first column is negative. There are two sign changes, indicating that the system has two RHP poles and one LHP pole, which makes the system unstable.
- If $K = 1386$, we have an entire row of zeros, which could signify $j\omega$ poles. Returning to the s^2 row and replacing K with 1386

$$P(s) = 18s^2 + 1386$$

$$\frac{dP(s)}{ds} = 36s$$

s^3	1	77
s^2	18	1386
s^1	~~0~~ 36	
s^0	1386	

Since there are no sign changes from the even polynomial (s^2 row) down to the bottom of the table, the even polynomial has its two roots on the $j\omega$-axis of unit multiplicity.

Since there are no sign changes above the even polynomial, the remaining root is in the left half-plane. Therefore the system is marginally stable.